Round 17 - Gaussian Hypertree Quartic Bank (GHQ-6)

A six-layer, all-distinct quartic source bank reusing the existing K3 propagator legs

[Derived design] Reuse each K3 source's existing dense legs P_s and A_s, but add a static symmetric birth-index core H_s. The quartic source is the sum of the leading connected Gaussian-tree star and path topologies. The dense pair coupling H_s makes the path term genuinely four-distinct; it is the missing topology that a column-separable "K4 riding K3 legs" cannot express. Keep six ages. No new dense transport is required.

1. State, birth, transport and readout

[Derived] Response coefficients at ReLU layer b. With sigma_i^2=C_ii, alpha_i=mu_i/sigma_i, define

```
r1_i = Phi(alpha_i),    r2_i = phi(alpha_i)/sigma_i,    r3_i = -alpha_i phi(alpha_i)/sigma_i^2.
```

[Derived] Birth objects. Let C0 be C with zero diagonal. Reuse the chain birth leg A_b = diag(r1) C0 and P_b=I. Add only

```
c_b = r3  in R^n,        H_b = diag(r2) C0 diag(r2)  in Sym_n.
```

For columns p_r=P[:,r], a_r=A[:,r], define the carried quartic source

```
Q_b = 4 Sym4 sum_r c_r (p_r tensor a_r tensor a_r tensor a_r)
 + 12 Sym4 sum_{r,t} H_{rt} (p_r tensor a_r tensor p_t tensor a_t).
```

[Derived] The coefficients 4 and 12 are the counts of labelled star and path trees on four vertices. Their absolute level is not relied on: the final diagonal is rescaled to the chain's per-layer K4 level, as required by the prompt.

[Derived] Genuine all-distinct birth. For four distinct indices i,j,k,l, the star part contains terms such as r3_i r1_j r1_k r1_l C_ij C_ik C_il. The path part contains all 12 chains, e.g. r1_i r2_j r2_k r1_l C_ij C_jk C_kl. Thus Q_b has non-zero (1,1,1,1) entries at birth; repeated labels also give non-zero (2,1,1) entries.

[Derived] Transport. Use exactly the same linearized map U as the K3 source bank:

```
P_s <- U P_s,      A_s <- U A_s,      c_s and H_s remain fixed in birth coordinates.
```

Therefore Q_s <- U^{tensor 4} Q_s exactly within this representation. The cheap carrier keeps ages 1-6; no rank truncation is used.

[Derived] Diagonal readout. For X_s=P_s o A_s,

```
q_star = 4 (P_s o A_s o A_s o A_s) c_s,
q_path[i] = 12 X_s[i,:] H_s X_s[i,:]^T,        q4_GHQ = sum_s (q_star + q_path).
```

On L12-L16, replace only the K4 pattern in the existing mean closure and rescale it to the chain's K4 level. Mean, C and the K3 chain are otherwise unchanged.

2. Why the previous kills do not cover this carrier

- **[Quoted]** The 2-3-layer young K4 carrier was tested with a horizon shorter than the measured six-layer memory (corr 0.77 at h=6). GHQ-6 changes that exact limitation: six ages are retained.
- **[Derived]** The earlier "K4 riding K3 source legs" was column-separable. Such a rider can make star-like one-latent-index terms but has no dense coupling between two birth indices. GHQ-6 adds H_rt and therefore the path topology; this is a strictly larger quartic subspace.
- **[Derived]** This is not a matrix/slice state: H is a fixed birth-coordinate coupling embedded inside four transported legs. The current K22/K31 slices do not determine it, and the represented tensor contains independent four-distinct entries.
- **[Derived]** This is not CP/Tucker/TT compression. H is full rank and untruncated in the ceiling test; the only cheap truncation is age=6, justified by the measured memory horizon.
- **[Inference]** The full true one-step linear K4 carry reaches 0.77-0.80, while fresh Gaussian physics is only 1-4% deep. GHQ-6 preserves old star/path sources exactly under U^{tensor4}; the ceiling test asks whether those two leading connected topologies span enough of the persistent pattern. There is no claimed measured correlation yet.

3. Exact cost at n=1024

item	real shape / count	cost
extra transport	none; P,A already carried by K3	0 u
birth H	15 births x 2 n^2 multiplies	[Derived] 0.0147 u total
deep path products	6 sources x 5 targets (L12-L16) = 30 square products	[Derived] 30 u classical
scored mixed schedule	L12: 6 classical; L13-L16: 24 products at quoted 0.51 u	[Derived] 18.24 u
all O(n^2) readout work	30 source-targets; 8 n^2 - 2 n FLOPs each	[Derived] 0.1172 u
TOTAL extra	mixed schedule + births + O(n^2)	[Derived] 18.37 u
extra state memory	6 full H matrices, float32	[Derived] 24.0 MiB + negligible vectors

[Estimated metered operations] Batch the six source products into one Strassen family per target layer. Use one classical batched matmul at L12 and four batched L5 families at L13-L16. Using the previously measured 110-240 operations per L5 family gives about 441-961 large-product operations, before a small number of fused elementwise calls. This is an estimate, not a measurement; if batching does not preserve that operation count, the residual cap may become the practical limiter.

4. Ceiling-first falsifier (self-contained script)

[Provided, not run here] The companion script round17_ghq6_falsifier.py is self-contained and uses only the paths in the prompt. It performs two tests:

- **[Derived test A]** Exact untruncated GHQ ceiling: retain every birth from layer 1, with full H and no age/rank truncation. Compare its K4 diagonal pattern with the stored 6e7 teacher at L12/L15/L16.
- **[Derived test B]** Cheap GHQ-6: same carrier but keep only source ages <=6. No fitting to K4d is performed. The script also runs an independent true-MLP Monte Carlo check with N=2,000,000 by default (float32 forward, float64 moment accumulation).

[Pre-registered gate] CEILING PASS if all-history corr >=0.70 at L12/L15/L16. CHEAP PASS if age<=6 also clears 0.70 at all three. KILL if all-history corr <0.55 at all three. Anything else is MIXED and should not be tuned before inspecting the topology residual.

[Quoted target] Correlations are computed only on decisive neurons |mu/sigma|<1.5. The stored 6e7 teacher is the primary gate because it has higher S/N; the fresh 2e6 Monte Carlo result is printed as an independent sanity check.

[Implementation note] The script reconstructs P and A itself from teacher means/covariances and the true weights, so it does not depend on estimator internals or r7 source captures. This makes the ceiling test independent of the current chain's source approximations.

Expected output format:

```
L12: all-history_vs_6e7=...  age<=6_vs_6e7=...  all-vs-MC2e6=...
L15: ...
L16: ...
PRE-REGISTERED VERDICT: PASS / CEILING PASS-CHEAP FAIL / KILL / MIXED
```

Bottom line: GHQ-6 is the cheapest factorised carrier I can identify that (i) has genuine (2,1,1) and (1,1,1,1) content at birth, (ii) preserves six-layer memory, (iii) reuses the expensive K3 transport, and (iv) fits the <=30 u envelope. Its only expensive new operation is the dense pair-coupled path readout. The representation should be killed immediately if the all-history script ceiling is below 0.55 at all three deep layers.